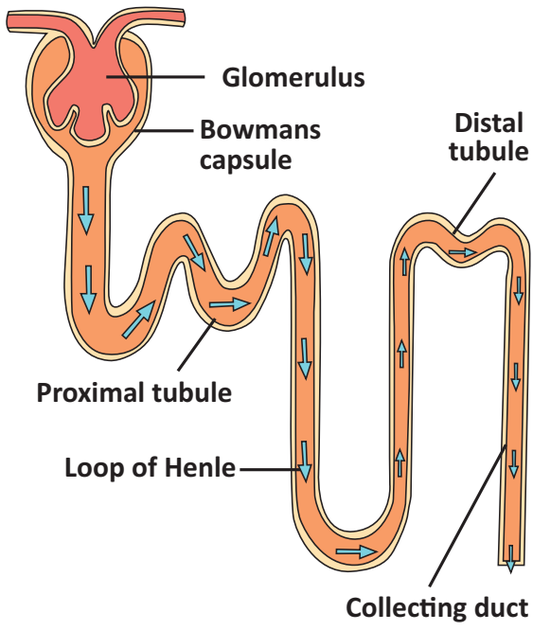


Tubules consist of different segments: the proximal convoluted tubule, loop of Henle, distal tubule and collecting duct. Each nephron functions independently and contributes to the final urine.

The glomerulus is a spherical mass of capillaries surrounded by a membrane, Bowman's capsule. The hydrostatic pressure within the glomerular capillaries results in the filtration of fluid into Bowman's capsule. The filtrate formed in the glomerulus passes first into the proximal convoluted tubule and from there, through the loop of Henle and distal convoluted tubule to the collecting ducts.

The processes of selective reabsorption and tubular secretion of the filtrate results in the production of 1.5 litres of urine that is excreted each day.

DO YOU KNOW?

The glomerular filtration rate (GFR) is used to monitor kidney function. Normal GFR is 100-125 ml/minute/1.73m^2. Chronic Kidney Disease occurs when GFR is < 60 ml/minute/1.73m^2 for 3 or more months.

Functions of the Kidney

The kidneys perform the following functions :

1. **Urine formation**-Make urine through which they excrete most of the waste products of metabolism like urea, creatinine, toxins and drugs.
2. **Maintain fluid and electrolyte balance**- Controls the concentration of the constituents of the blood like sodium, potassium, chloride etc.
3. Kidneys are an **exclusive site for the production** of

a) 1,25 dihydroxy cholecalciferol, the active from of vitamin D which is important for bone health.

b) Erythropoietin which is essential for the synthesis of red blood cells.

c) Renin, an enzyme that helps to regulate blood pressure.

4. **Hormone degradation** - Parathyroid hormone, calcitonin, insulin and gastrin are degraded by the kidneys.

Risk factors for kidney disease

- Old age
- Exposure to hot climate
- Diabetes
- Hypertension
- Urinary tract infections
- Urinary stones
- Family history of chronic kidney failure
- Exposure to certain nephrotoxic drugs
- Low birth weight

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